

Software Requirements Specification document for TBS

Submitted by
Rohit Suresh

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INTRODUCTION

A tour booking system is a digital platform that allows customers to browse, book, and manage travel-related services such as tours, activities, vacation packages, hotel stays, transportation, and sightseeing experiences. These systems are commonly used by travel agencies, tour operators, and tourism businesses to automate reservations and improve customer service.

The main purpose of a tour booking system is to simplify the booking process for both customers and administrators. Customers can search for available tours, compare prices, view schedules, make secure online payments, and receive instant confirmations. On the business side, the system helps manage bookings, customer information, availability, payments, cancellations, and reporting from a centralized platform.

Clearly defining system requirements is one of the most important steps in the successful development and implementation of a tour booking system. System requirements describe what the system should do, how it should perform, and the constraints under which it must operate. Well-defined requirements act as a foundation for the entire project.

The importance of clearly defining requirements includes:

- **Improved Understanding of Business Needs:**
Requirements help developers, designers, and stakeholders understand the goals of the system, such as online booking, payment processing, customer management, and reporting.
- **Reduced Development Errors:**
Clear requirements minimize misunderstandings and confusion during development, reducing the chances of building incorrect or incomplete features.
- **Better Project Planning:**
Accurate requirements allow teams to estimate costs, timelines, resources, and technical needs more effectively.

- **Enhanced System Quality:**

Defining functional and non-functional requirements ensures the system is reliable, secure, user-friendly, and capable of handling expected workloads.

- **Efficient Communication:**

Requirements provide a common reference point for clients, developers, testers, and project managers, improving coordination throughout the project lifecycle.

- **Simplified Testing and Maintenance:**

Clearly documented requirements make it easier to verify whether the system meets expectations and to maintain or upgrade the system in the future.

- **Reduced Risk and Cost Overruns:**

Identifying requirements early helps avoid major changes later in development, saving time and reducing additional costs.

In summary, clearly defining system requirements ensures that the tour booking system is developed according to user needs, delivered on time and within budget, and functions effectively after implementation.

PURPOSE

The purpose of the Software Requirements Specification (SRS) document for the tour booking system is to provide a clear, detailed, and organized description of the system's functional and non-functional requirements. It serves as an official reference for developers, designers, testers, project managers, and stakeholders throughout the software development lifecycle.

The SRS document aims to:

- Define the features and functionalities of the booking system, such as tour search, online reservations, payment processing, customer management, and booking confirmations.
- Specify system performance, security, usability, and reliability requirements.
- Establish a common understanding between clients and the development team regarding system expectations and project scope.
- Serve as a foundation for system design, development, testing, and maintenance.
- Reduce misunderstandings, development risks, and unnecessary changes during implementation.
- Provide criteria for validating and verifying that the completed system meets user and business requirements.

In summary, the SRS document ensures that the booking system is developed systematically, efficiently, and according to the agreed business objectives and user needs.

SCOPE

The scope of the tour booking system defines the boundaries, features, and functionalities that the system will include to support tour reservation and management processes for customers and administrators.

The system is designed to provide an online platform where users can browse available tours, view tour details, make reservations, and complete secure payments. It also enables administrators and tour operators to manage tour packages, schedules, customer bookings, payments, and reports efficiently.

Scope of the System

The tour booking system will include the following functions:

- User registration and login
- Tour package listing and search functionality
- Viewing tour details, prices, schedules, and availability
- Online booking and reservation management
- Secure payment processing
- Booking confirmation and notification services
- Customer profile and booking history management
- Admin panel for managing tours, customers, and bookings
- Report generation and analytics
- Cancellation and refund management

Users of the System

- Customers/Tourists
- Consultants
- Agency Admin

System Boundaries

The system focuses on managing tour booking activities and related customer interactions. It does not include services such as direct airline operations, visa processing, or external transportation management unless integrated through third-party services.

In summary, the scope of the tour booking system is to automate and simplify the process of tour reservation, management, and customer service while improving operational efficiency for tourism businesses.

OBJECTIVES

High-Level Objectives of the Tour Booking System

The high-level objectives of the tour booking system are to improve the efficiency, accessibility, and management of tour reservation services for both customers and tour operators. The system aims to automate booking operations and enhance the overall user experience.

The main objectives include:

- 1. Provide Online Tour Booking Services**

Enable customers to browse, select, and book tour packages anytime and from anywhere through an online platform.

- 2. Automate Booking and Management Processes**

Reduce manual work by automating reservations, payment processing, confirmations, cancellations, and scheduling.

- 3. Enhance Customer Experience**

Offer a user-friendly interface with quick access to tour information, secure payments, and instant booking confirmations.

- 4. Improve Operational Efficiency**

Help administrators and tour operators manage tours, customers, bookings, and reports more effectively from a centralized system.

5. Ensure Secure and Reliable Transactions

Protect customer data and payment information through secure authentication and payment systems.

6. Provide Accurate Information and Reporting

Generate real-time reports and maintain accurate booking records to support business decision-making.

7. Support Scalability and Future Expansion

Design the system to accommodate increasing users, additional tour packages, and future feature enhancements.

Overall, the booking system aims to streamline tour reservation activities, improve service quality, and support the growth of tourism businesses.

The tour booking system is designed to address several challenges faced by customers and tour operators in managing tour reservations and travel services. Its primary goal is to automate and simplify the booking process while improving efficiency, accuracy, and customer satisfaction.

Goals and Problems the System Aims to Address

1. Reduce Manual Booking Processes

Traditional booking methods often involve paperwork, phone calls, and manual record-keeping, which are time-consuming and error-prone. The system automates these processes to save time and reduce human errors.

2. Improve Customer Convenience

Customers may face difficulties accessing tour information, checking availability, or making bookings outside business hours. The system provides 24/7 online access to tours, schedules, pricing, and reservations.

3. Eliminate Booking Errors and Double Reservations

Manual systems can lead to incorrect bookings or duplicate reservations. The system uses real-time availability tracking to ensure accurate and reliable booking management.

4. Enhance Booking Management Efficiency

Tour operators often struggle to manage customer records, payments, cancellations, and schedules efficiently. The system centralizes these operations in one platform for easier administration.

5. Provide Secure Payment Processing

Handling payments manually may create security risks and delays. The system supports secure online payment methods to improve transaction safety and speed.

6. Improve Communication and Notifications

Customers may not receive timely updates about bookings or cancellations. The system sends automated confirmations, reminders, and notifications to keep users informed.

7. Support Business Growth and Decision-Making

Without proper reporting tools, businesses may find it difficult to analyze performance and customer trends. The system generates reports and analytics to support informed decision-making.

8. Increase Service Quality and Customer Satisfaction

By offering faster bookings, accurate information, and better customer support, the system improves the overall travel booking experience.

In summary, the tour booking system aims to solve inefficiencies in traditional booking methods by providing a secure, automated, and user-friendly platform for managing tour reservations and related services.

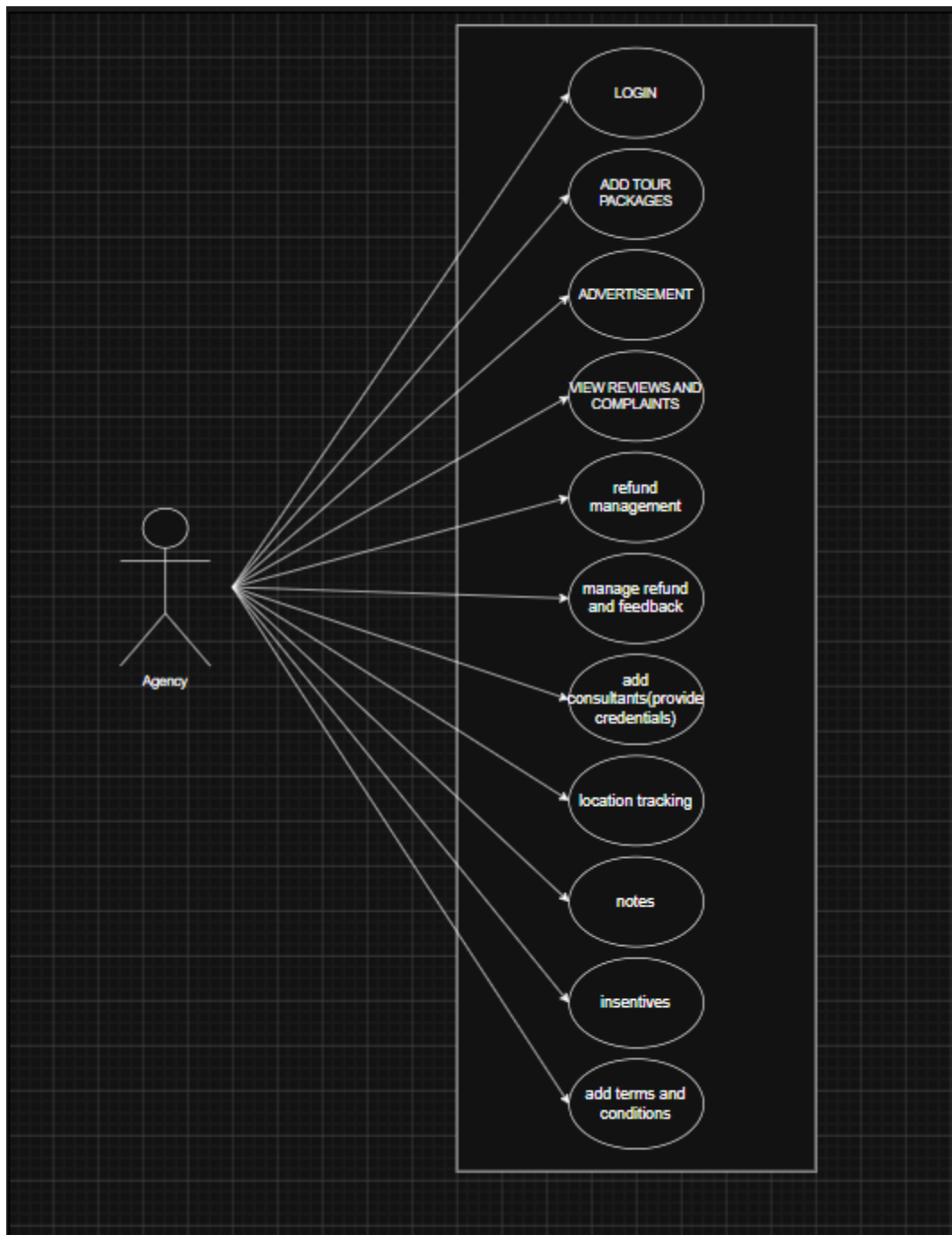
FUNCTIONAL REQUIREMENTS

Primary actors of the system

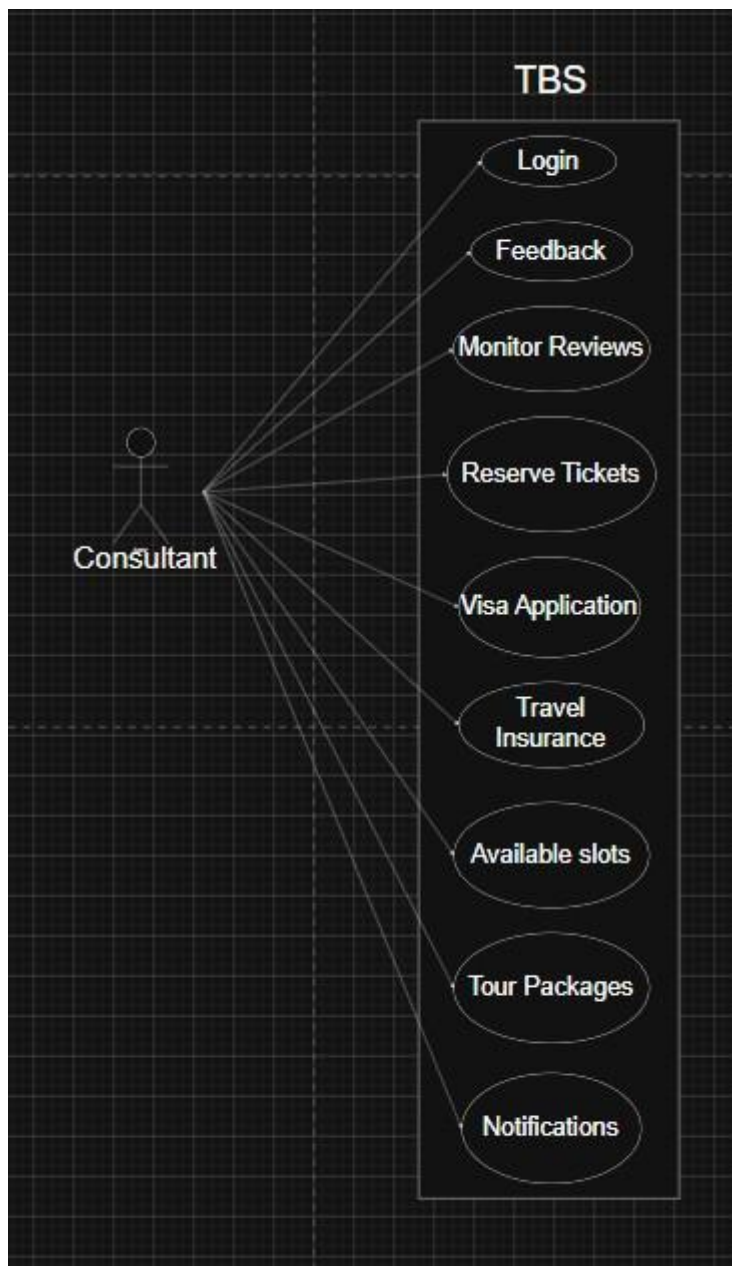
- 1) Agency
- 2) Consultant
- 3) Customer

Use Case Diagrams

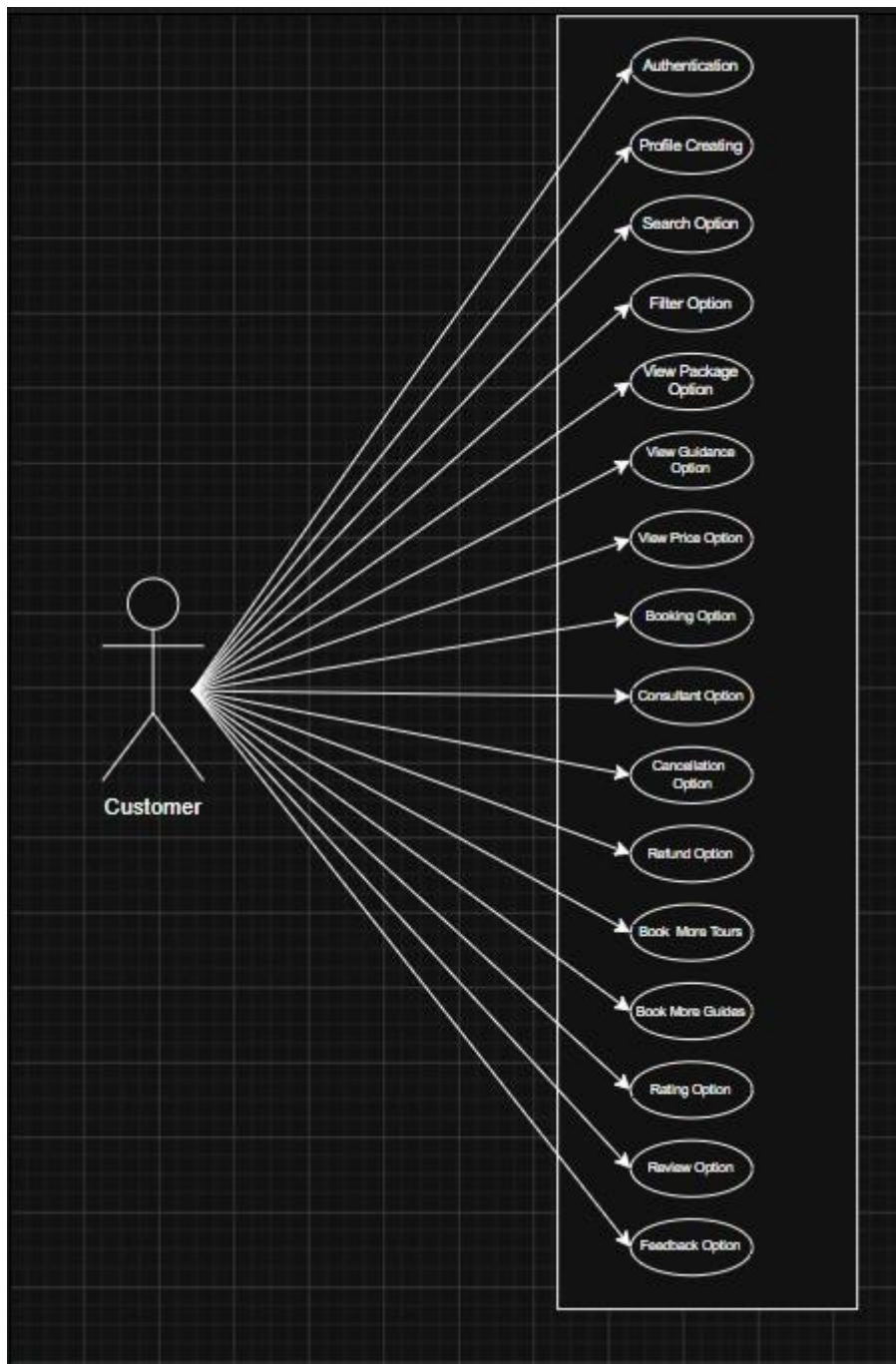
- 1) Agency



2) Consultant



3) Customer



User Stories

1) Agency

As a (user role)	I want (goal)	So that (benefit)
AGENCY	I want to login	So that I can manage the entire process
AGENCY	I want to add tour packages	So that I can provide details of tour packages
AGENCY	I want to give advertisement	So that I can promote this tour agency
AGENCY	I want to view reviews and complaints	So that I can improve the quality of service
AGENCY	I want to manage refund activities	So that I can satisfy the customer and improve customer trust
AGENCY	I want to manage feedback	So that I can delete any bad feedbacks
AGENCY	I want to add consultants(provide credentials)	So that they can login and work under the agency and manage consultant activities
AGENCY	I want to track the location	So that I can know the current location of customers and ensure safe and efficient travel
AGENCY	I want to add notes	So that I can provide additional information to the customer
AGENCY	I want to give incentive to the consultants	So that I can encourage consultants to work efficiently
AGENCY	I want to add terms and conditions	so that they can know the rules for using this web application

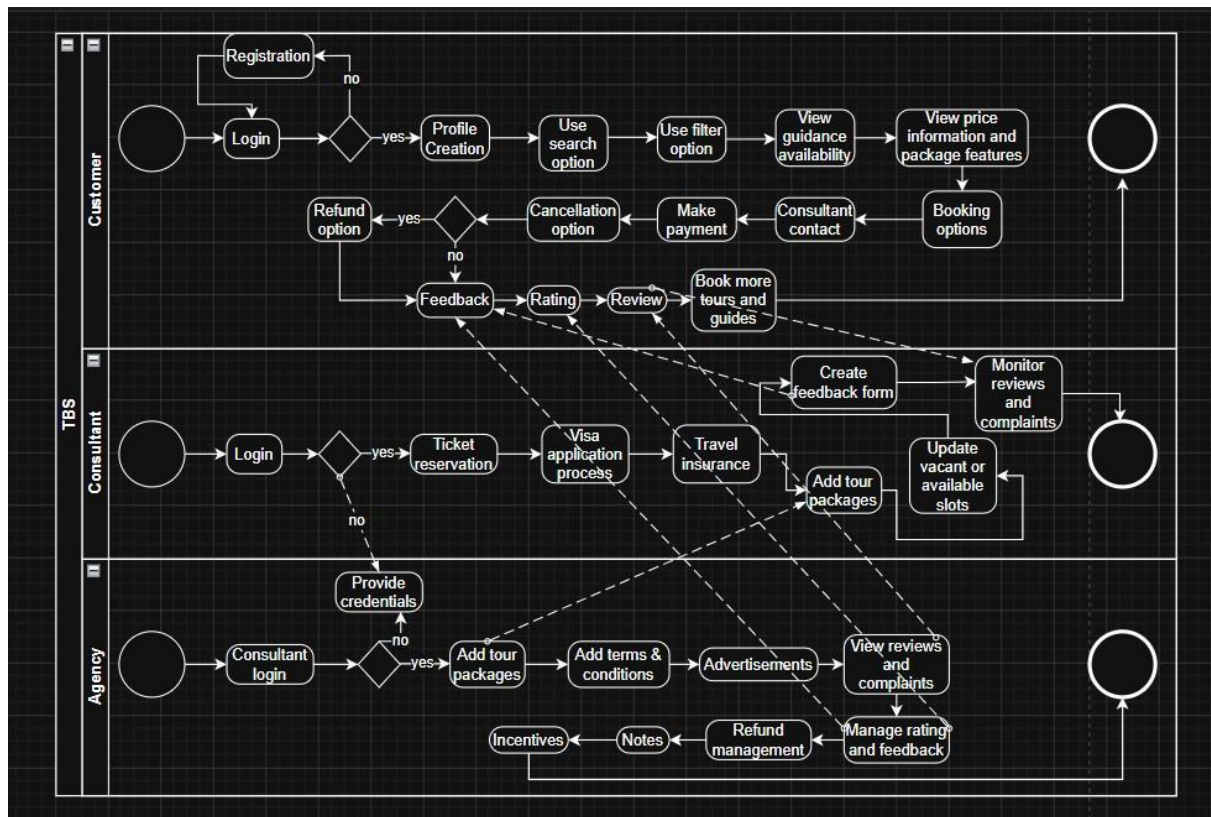
2) Consultant

As a	I need	So that
Consultant	To allow users to register and log in to the application	Customers can safely access their accounts and manage bookings easily
Consultant	To create a feedback form for customers	Customers can share their opinions and help improve the service
Consultant	To monitor reviews and complaints	Customer problems can be solved quickly and service quality can improve
Consultant	To provide an option for reserving tickets	Customers can book tickets easily and save time
Consultant	To assist customers with the visa application process	Customers can get support and guidance for visa approval
Consultant	To provide travel insurance services	Customers can travel with financial protection and peace of mind
Consultant	To display tour vacancies and availability	Customers can quickly check available slots before booking
Consultant	To add and manage tour packages	Customers can choose tour packages that fit their needs and budget
Consultant	To send text and email notifications	Customers can receive booking updates and reminders on time

3) Customer

<i>As a (user role)</i>	<i>I want (goal)</i>	<i>so that (benefit)</i>
<i>Customer</i>	<i>Register into the application by entering details.</i>	<i>Make my data available to agency and use the system capabilities reserved for registered users.</i>
<i>Customer</i>	<i>I want to create my profile with my interests</i>	<i>So that the agency can easily provide tour details to me that can match my interests.</i>
<i>Customer</i>	<i>I want a search option</i>	<i>So that I can quickly discover tours worldwide without browsing endlessly.</i>
<i>Customer</i>	<i>I want a view packages feature</i>	<i>So that I get the best value and avoid hidden surprises.</i>
<i>Customer</i>	<i>I want view guidance availability</i>	<i>So that I can choose tours with expert guides, enhancing my experience with local insights and cultural knowledge.</i>
<i>Customer</i>	<i>I want to view price information</i>	<i>So that I can make budget effectively, and avoid unexpected costs.</i>
<i>Customer</i>	<i>I want a booking option</i>	<i>So that I can secure my spot instantly and avoid missing out on popular tours.</i>
<i>Customer</i>	<i>I want consultant contact option</i>	<i>So that I can get expert to clarify doubts and feel confident before committing to a tour.</i>
<i>Customer</i>	<i>I want a cancellation option</i>	<i>So that I can cancel my tour if necessary.</i>
<i>Customer</i>	<i>I want a refund option</i>	<i>So that I don't lose money when cancelling, making the service trustworthy and risk-free.</i>
<i>Customer</i>	<i>I want a book more tours</i>	<i>So that I can easily extend my travel</i>
	<i>/guides option</i>	<i>experience, bundle multiple tours and customise my tour with additional expert guides if necessary.</i>
<i>Customer</i>	<i>I want a rating, review & feedback option</i>	<i>So that I can share my opinion, document my experience in details to contribute to improving tour quality and help the future travellers make better choices.</i>

BUSINESS PROCESS DIAGRAM



Non-Functional Requirements

Non-functional requirements for a PC (Personal Computer) describe the quality attributes, performance standards, and operational conditions the system should meet rather than specific functions it performs.

Non-Functional Requirements of a PC

1. Performance

- The PC should boot within an acceptable time (e.g., under 30 seconds).
- Applications should respond quickly without significant delays.
- The system should support multitasking efficiently.

2. Reliability

- The PC should operate continuously without frequent crashes or failures.
- Data should remain stable and protected during normal operation.

3. Usability

- The system should provide an easy-to-use interface.
- Users should be able to access applications and settings without difficulty.

4. Security

- The PC should support password protection and user authentication.
- Antivirus and firewall protection should be available to prevent unauthorized access and malware attacks.

5. Scalability

- Hardware components such as RAM, storage, and graphics cards should be upgradeable.
- The system should support installation of additional software and peripherals.

6. Compatibility

- The PC should support commonly used operating systems, software applications, and external devices.
- It should be compatible with printers, USB devices, monitors, and network systems.

7. Maintainability

- The system should allow easy troubleshooting, repair, and software updates.

- Hardware replacement and maintenance should be manageable.

8. **Availability**

- The PC should remain accessible and operational whenever needed, with minimal downtime.

9. **Portability** *(for laptops)*

- The device should be lightweight and easy to carry.
- Battery life should support extended usage.

10. **Efficiency**

- The PC should use system resources such as CPU, memory, and power efficiently.

11. **Backup and Recovery**

- The system should support data backup and recovery mechanisms to prevent data loss.

These non-functional requirements help ensure that the PC performs efficiently, securely, and reliably while providing a good user experience.

TECHNICAL REQUIREMENTS

Device info:

Device name	Dell Inspiron 15-3567
Processor	Intel(R) Core(TM) i7-7500U CPU @ 2.70GHz (2.90 GHz)
Installed RAM	8.00 GB (7.87 GB usable)

Graphics card	Radeon (TM) R5 M430 (2 GB) Intel(R) HD Graphics 620 (128 MB)
Storage	954 GB
System type	64-bit operating system, x64-based processor

Windows info:

Edition	Windows 11 Home Single Language
Version	25H2
OS build	26200.8457
Experience	Windows Feature Experience Pack 1000.26100.304.0

Programming languages:

- 1)HTML
- 2)CSS
- 3)JAVASCRIPT
- 4)PYTHON

Frameworks:

REACT (library)

FASTAPI

Platforms:

- 1)Trello
- 2)Drawio

CONCLUSION

The Tour Booking System is designed to provide an efficient, secure, and user-friendly platform for managing tour reservations and related travel services. The system automates key booking activities such as tour searching, reservation processing, payment handling, customer management, and report generation. By replacing manual processes with a centralized digital solution, the system improves operational efficiency, reduces errors, and enhances customer satisfaction.

This Software Requirements Specification (SRS) document has defined the purpose, scope, objectives, functional requirements, and non-functional requirements of the system. The document serves as a foundation for the design, development, testing, and maintenance of the application. Clearly identifying system requirements ensures that all stakeholders share a common understanding of the system's goals and expected behaviour.

The proposed system aims to solve common challenges in traditional booking methods, including manual record management, booking conflicts, delayed communication, and inefficient customer service. Through automation and real-time data management, the system supports better decision-making, reliable service delivery, and future scalability.

Closing Remarks / Recommendations

To ensure the successful implementation of the Tour Booking System, the following recommendations should be considered:

- Develop the system using a scalable and secure architecture to support future enhancements and increasing user demand.

- Ensure strong security measures are implemented to protect customer data and online transactions.
- Design a simple and responsive user interface to improve accessibility and user experience.
- Conduct thorough system testing to verify that all functional and non-functional requirements are satisfied.
- Provide regular system maintenance, updates, and technical support to ensure long-term reliability and performance.
- Consider integrating additional features in the future, such as mobile applications, multilingual support, online reviews, and third-party travel service integrations.

In conclusion, the Tour Booking System will provide a modern and reliable solution for managing tour reservations efficiently while improving customer convenience and supporting business growth in the tourism industry.